More Than a Game: Victory-Driven Crime Spikes and the Complex Urban Impact of Sports

Evan McDermid[a] **and Paul Waxman**[a]

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Major sports events generate strong emotions and large crowds, creating conditions that can have large influence on crime patterns in surrounding areas. In this paper, we build on research connecting emotion, alcohol consumption and large groups to violence, investigating the different ways that professional sports games can affect urban crime in real time. We analyse over 10 years' worth of location-based crime data from Philadelphia and Chicago, aligning daily crime rates with game dates for the Eagles, 76ers and the Bears. Using game schedules and outcomes, we calculate the changes in crime on game days relative to non-game days. We examine how distance affects this by computing concentric bands around each stadium. Using Welch's t-tests, we look at the statistical significance of crime differences by game outcomes. We find that crime increases near stadiums on game days, especially within 3km of the stadium following team victories. However, we also find that crime stays about equal as a whole on game days; the increase only occurs around the stadium. These patterns are more pronounced in Philadelphia than in Chicago. Another key finding is that crime rates are lower in cities on days of major sporting events, however, we see a large spike the day after if the team wins. These findings contribute to research on how collective emotions can manifest in public behaviour, and they suggest the need for specific policing strategies during and after high-profile games. More broadly, the results highlight how sports can measurably impact urban dynamics.

urban crime | sports events | crowd behavior | policing | win-loss effects

Introduction / Related Work

Major sporting events are very emotionally charged and widely attended, which has been seen to often trigger shifts in public behaviour, in both good and bad ways. Previous research has documented increases in domestic violence after international football matches that England play in, particularly after losses (1). Recently, the media has even reported how victories can lead to unrest and crime spikes, such as the riots in Paris after PSG won the Champions league(2).

Initially, we were looking at how crime rates in the UK are affected by Premier League and England matches, but the UK government crime data did not include timestamps, so it made robust analysis difficult. We therefore shifted focus to the U.S., where detailed open databases were available for cities Philadelphia and Chicago, which both are big sports cities.

We are interested in understanding the different mechanisms by which sports events can affect crime. We think these could include emotion, alcohol, shifts in police activity, or just sheer crowd movement and increased population density.

Methodologically, this project aligns crime data with team schedules and results, analyses crime changes on game days, computes spatial proximity of crimes to stadiums, compares cities, teams, sports and game outcomes, and applies statistical testing to assess significance.

This builds on the prior research by contributing a spatial view of crime near stadiums, a comparative perspective across time, cities and sports, and a methodology that pairs formal tests with visual tools.

Data

Data Sources. Crime data for this study was collected from open data crime portals of two major U.S. cities: **Philadelphia**, using the Crime Incidents dataset from https://data.phila.gov, and **Chicago**, using crime reports from the Chicago Data Portal at https://data.cityofchicago.org. These two cities were chosen as they have prominent sports teams, and the crime data included specific timestamps.

Time Window. We focus on the years 2020–2024 for both cities, but include years as early as 2008 to align with periods of significant local sports events and NFL and NBA data availability.

Author affiliations: [a] Dartmouth College

Original Units of Analysis. Each row in the data corresponds to a single police-reported crime event and includes columns such as timestamp, type of crime, and geographic coordinates.

Our Analytical Units. Depending on the task, we restructure the citywide crime data in a couple of ways: (1) daily crime counts citywide, and (2) Δ crime by distance bands from stadiums.

Missingness and Filtering. Rows missing key geographic fields (latitude, longitude) or valid date entries were dropped.

Limitations. Our analysis is dependent on and subject to possible geocoding inaccuracies near city boundaries, variation in reporting practices across neighbourhoods and especially over time, and jurisdictional differences—particularly near stadiums.

Code Reference. See `code/01_clean_and_tag_crime.py`, lines 22–45, for preprocessing logic and filtering implementation.

Methods

Our analysis followed six major sections: data cleaning, game tagging, distance based grouping of crime, computation of crime deltas, event centered analysis, and win-loss analysis. All code used in this section is available and can be found in the GitHub repository, and it is organised by stage and ordered by number of executions.

Data Cleaning. We used crime datasets available on the Philadelphia Open Data Portal and the Chicago Data Portal, focusing on the years 2020-2025 but covering years as early as 2008 for specific event analysis. Each city's dataset contained crime reports and included crime types, geographic coordinates and important timestamps. We standardised column names across the two datasets within the cleaning and made sure the date column was parsed to datetime to explicitly specify the format to avoid inconsistencies. Timestamps were filtered to the nearest day using `dt.floor("D")` to look at daily aggregation.

We filtered out any rows including missing values from fields such as latitude and longitude, or timestamp. Latitude and longitude are put into numeric format, and rows with missing or invalid coordinates are dropped. The cleaned dataset was sorted with consistent names of `date`, `lat`, `lng`.

In addition to cleaning the crime data, we also collected NFL game data to be able to tag crime incidents relative to game days. Using custom function `nfl_game_date()`, we extracted and cleaned the game data from a public Kaggle dataset.

For each season we downloaded the CSV containing home and away games and extracted only rows weather the specific team played and constructed the Date column for each game. Due to NFL seasons crossing years we adjusted the date formatting by inferring the year based on the month. Each game was tagged with a location, Home or Away, and result, Win or Loss. This processed game schedule was later used to define which crimes where on game days and compare the patterns. See `code/01_load_game_data.py`, lines 5–40.

Game Tagging. To identify which crimes happened on game days we made a custom tagging function `tag_game_windows`. This function merged the crime dataset with a dataset of home game dates for the specific team we were looking at. Each crime report was assigned a binary column `is_game_window` indicating whether it occurred on one of the home game days or not. We also gave each relevant row the corresponding game result where it was applicable. This tagging method allowed us to isolate crime on game days for further comparisons.

Distance Binning. To capture how crime rates changed based on distance away from the stadium in question, we computed the distance from each crime report to the home stadium using a formula implemented via the `geopy` and `shapely` libraries. The coordinates for the home stadium were fixed previously

Crimes were grouped into distance bands based on how far they were from the stadium. We settled on using bins of [0–1 km], [1–5 km], [5–10 km], and [10km +] to see differences between near, mid-range and city-wide effects. These bins were implemented using `pd.cut` over the pre-computed distances in meters.

Game-Day Δ Crime Calculation. For each game day we calculated the difference in total crime counts compared to the average crimes on non-game days in the same season. To be more specific, for each distance grouping, we computed the average number of crimes per day on a non game day and subtracted it from the number calculated on a game day, normalised to the number of days for each. This gave us a Δ crime score per game per distance.

This approach controlled underlying seasonal crime variations that might exist and allowed us to focus on the effect on just game days. See `code/03_analyze_delta_by_distance.py`, lines 10–40, to see how it was implemented.

Event-Centered Analysis. As well as looking at game days in general, we also analysed several high-profile specific events such as Super Bowls or World Series by comparing the crime levels on the day of the event and the day after to the average crime rate from the year before.

We used the function `compute_event_and_followup_crime_delta` to calculate the number of crimes on the event date as well as the following day, and the average over the previous year. This allowed us to see both the immediate and delayed effects of large-scale events in cities. Maps were also created to show where crimes were concentrated on the two days in focus

Win-Loss Analysis. Finally, to assess whether game outcomes significantly influenced crime levels, we conducted Welch's t-tests to compare crime counts on win and loss game days. This was done on a city-wide level as well as filtering to a 3km stadium radius using `scipy.stats.ttest_ind` with an assumption of unequal variance.

Results

Crime Changes on Regular Game Days (Eagles, Bears, Sixers). We begin by looking at how crime levels change on regular game days for three teams: the Philadelphia Eagles, Chicago Bears, and Philadelphia 76ers. The analysis of these three is split by location and by team.

For the Eagles we find that citywide crime actually decreases on gamedays, as shown in Figure 1. When we look at the trend as a whole we see a decrease in citywide crime. However, looking at the 1km radius around the stadium, the pattern reverses, and we see a crime delta of 1.3, which is a large change for such a small area.

These two patterns can be explained by a few contextual factors. Eagles games are predominantly played on Sundays, when crime rates in Philadelphia are already lower than average. Fans also may stay at home to watch the game, which would likely mean less crime on the streets. The concentration of fans, drinking, and movement around the stadium due to the 67000 capacity of the stadium likely explains the local spike, despite the rest of the city experiencing reduced crime rates.

In contrast, there is a citywide increase in gameday crime of about 4%. The largest percentage increase, however, also occurs within 1km of Soldier Field, a crime delta of 1.4 from a non-gameday baseline of 2.3 crimes in that area per day. This can be seen in Figure 1. The spike in crime around the stadium can be explained with the same reasons as above; the reason that the whole city sees an increase could be a few factors. Chicago is a very lively city, and people may go out to bars to watch the game as opposed to staying at home, which, due to alcohol and increased population, might result in increased crime. Another possibility is that the Chicago Bears lose more often, which could emotionally affect fans and cause increased crime for that reason.

The 76ers show no clear change in crime levels across the city or close to the stadium, as we see in Figure 1, which can be explained by three key reasons. There is much lower public engagement in basketball games, there are smaller crowds with the Wells Fargo Centre at a capacity of 20000, and each game is less important, so fans will be less emotionally charged.

These differences across team analysis suggest that sport, city and perceived importance are key factors in how games affect urban crime rates.

High-Profile Events and Post-Game Spikes. We next analyse specific high profile sports games involving Philadelphia and Chicago sports teams and compare crime levels on the day of the game versus the day after. In all of the cases of the teams winning, the most notable crime spikes occur the day after the event, not on the day of the event itself. This is what was seen with the PSG Champions League scenario, fans lined the streets to celebrate the victory the day after the event and ended up starting riots and committing crimes.

Sporting Event (Team)	$\triangle$ on Event Day	$\triangle$ on Next Day
	PHILADELPHIA	
2018 Super Bowl (Eagles)	-68.71	33.29
2024 Super Bowl (Eagles)	-118.79	73.21
2008 World Series (Phillies)	-118.87	106.13
2022 World Series (Phillies)*	0.49	-23.51
	CHICAGO	
2013 Stanley Cup Final (Blackhawks)	-16.83	70.17
2015 Stanley Cup Final (Blackhawks)	41.73	147.73
2016 World Series (Cubs)	12.44	47.44
2015 NLCS (Cubs)*	61.24	41.24

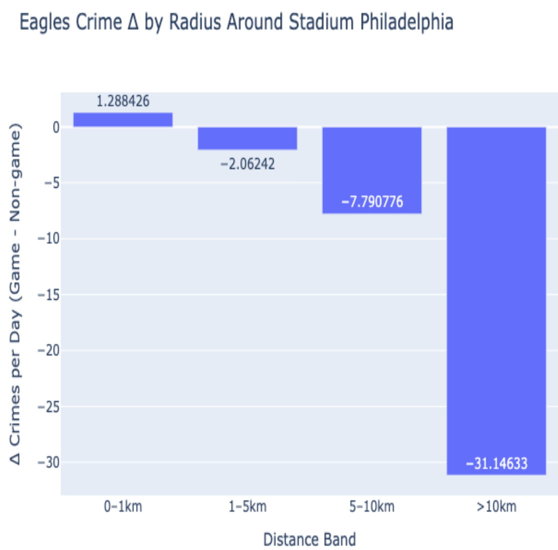
*These events are major losses

Table 1. Crime $\triangle$'s the Day of, and After Major Sporting Wins/Losses

In Philadelphia, maps of crime activity on and after the 2025 and 2019 Super Bowls, which they won, show large increases in crime, especially in the centre of the city, the day after the win. Figures S1b and S1c clearly show a shift from sparse crime dots (blue) on game day to red clusters the next day. Similarly, the 2008 Phillies World Series win shows a sharp post-event crime in central Philadelphia the next day. This can be easily reasoned, fans meet up in the centre of the city the day after the win to celebrate, and due to the high numbers, alcohol and emotion, there is an increase in crime in that area. This phenomenon is described in news reports which discuss arrests the day after major sporting wins.

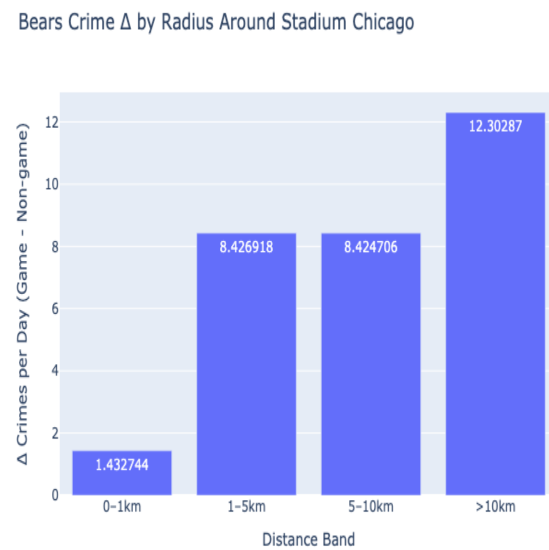
However, when we look at the 2022 Phillies, we see a different result. In 2022, the Phillies made it to the World Series but lost, and when we look at the crime rates, we see there is no spike in crime the following day, crime is dispersed as it usually is (Figure S1a). This shows that high-profile matches cause increases in crime the day after the games, but only in the city of the winning team.

Chicago follows a similar pattern. Both the 2016 Cubs World Series win and the 2015 Stanley Cup win show normal amounts of crime on the night of the event, followed by widespread red markers the day after. These markers are more spread out than in Philadelphia but show a similar trend, that celebrations in the aftermath are the primary drivers of increased crime citywide, more so than the game days themselves. It is notable that the delta for game day crime of these important matches



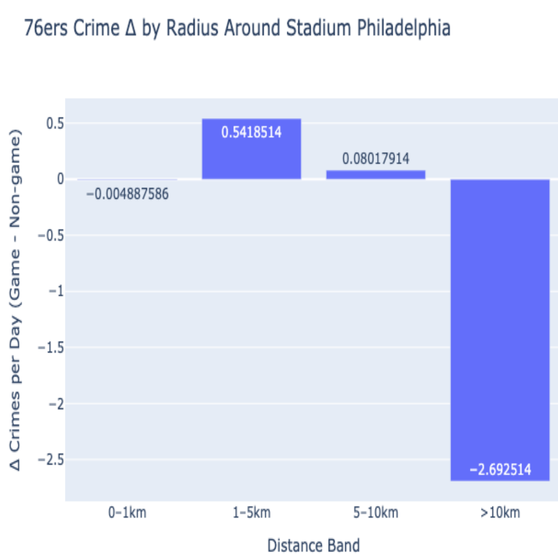
(a) Philadelphia Eagles

Crime decreases citywide on game days, especially in outer areas (¿10km), but shows a small increase near the stadium (0-1km).



(b) Chicago Bears

Crime increases across all distance bands on game days, with the largest spike occurring in areas ¿10km from the stadium.



(c) Philadelphia 76ers

Shows minimal change in crime patterns across all distance bands, indicating limited impact of basketball games on urban crime.

Fig. 1. Crime delta changes by distance from stadium for Philadelphia Eagles, Chicago Bears, and Philadelphia 76ers on regular game days. Each chart shows the change in crimes per day (Δ) compared to non-game days across different distance bands from the respective stadiums.

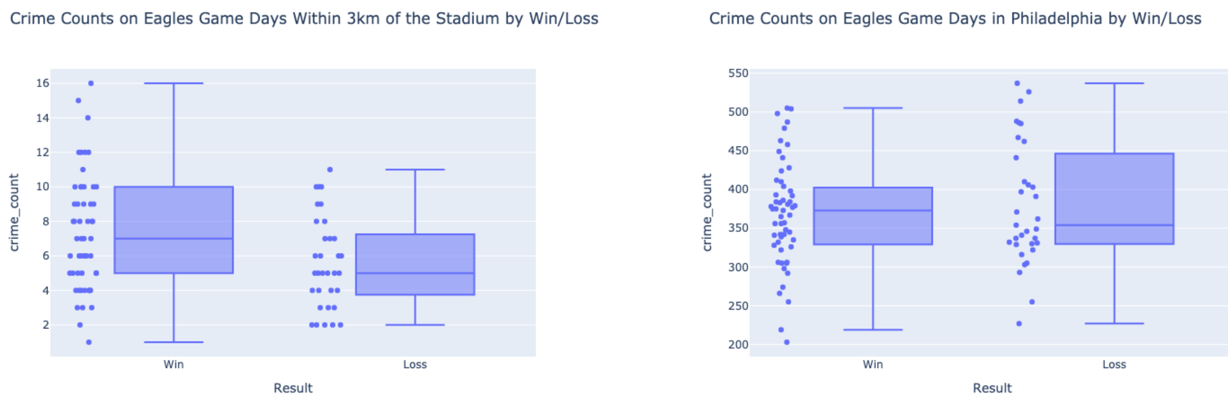
is much less negative in Chicago than in Philadelphia, reinforcing that people are more likely to be out watching the game in bars in Chicago and will therefore commit more crimes on game day.

Win vs. Loss Comparison: Statistical Testing. To examine how outcomes affect crime rates we group together game days by the team result and plot the distribution of crime counts. We created two box plots comparing these wins and losses, both as citywide crime counts and within 3 km of the stadium. We focused here only on Philadelphia, which revealed two key things:

Looking at the city as a whole, game day victories cause an increased median crime rate, however it is not significant. It more clearly shows that there is no large change in crime rates in the city based on whether the Eagles win or lose (Figure 2).

When we filter to only crimes within a 3km radius we see that there are many more crimes on win days. The Welch's t-test shown in **Table 2** shows that there is a statistically significant increase in crimes on win days than on loss days. The p value is 0.0085, which is much less than the threshold of 0.05 to show statistical significance.

This suggests a similar pattern to what we saw with analysis of important games, that celebratory behaviour, not frustration at losses, may be more crime-inducing in these contexts, particularly when paired with alcohol, large crowds and a police presence.



(a) Crime Near Stadium by Game Outcome

Crime counts within 3km of the stadium show higher median values and greater variability on Eagles win days compared to loss days.

(b) Citywide Crime by Game Outcome

Citywide crime counts reveal similar median levels between win and loss days.

Fig. 2. Crime count distributions on Eagles game days comparing wins versus losses. The left panel shows crime within 3km of the stadium, while the right panel shows citywide totals. Both reveal increased crime variability following victories, with celebration-related incidents contributing to higher outlier frequencies in win conditions.

Region	t-statistic	p-value	Significant?
Citywide (Philadelphia)	-0.74	0.4636	No
Within 3km of Stadium	2.70	0.0085	Yes

Table 2. Welch's t-test results comparing game-day crime after wins vs. losses in Philadelphia. Significance determined at $\alpha = 0.05$.

Discussion

This paper finds that major sporting events influence crime patterns in nuanced and location based ways. While overall crime often decreases in the city as a whole on gamedays, especially in Philadelphia, there is a consistent and significant increase in crime in the close vicinity of the stadium, which increases after team victories. High-profile events such as Super Bowls, World Series and Stanley Cups wins also trigger delayed spikes in crime, with the largest increases occurring the day after the event.

There are several important limitations to note. First there are confounding factors like weather, day of the week, or police deployment levels that are likely to affect crime types and counts but we did not directly control for. Second, our analysis aggregated all crime types, even though different crimes may be affected differently by game days. Third, some effect could be driven by non game related activities which are not controlled for, for example unrelated events near the stadiums.

Future research could address some of these issues. Incorporating arrest, emergency room, or 911 dispatch data could shed light on crime severity and types which would be interesting to further the analysis. Another valuable extension would be to integrate attendance data of games to directly see whether the population is a large factor in predicting crime spikes.

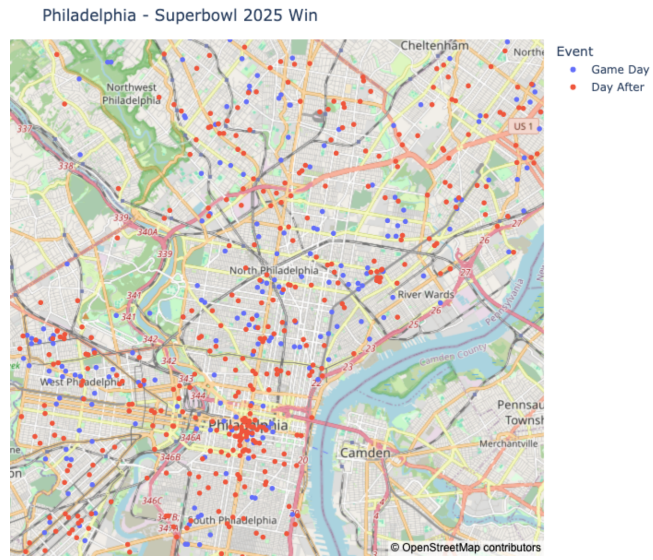
Overall, our findings emphasise that celebrations can increase crime rates more than disappointments, and that these effects are spatially concentrated around stadiums or city centres. Law enforcement may wish to target resources to these specific areas on game days and after high importance event victories, instead of just during the events themselves.

1. R Ivandić, T Kirchmaier, Y Saeidi, NT Blas, Football, alcohol, and domestic abuse. *J. Public Econ.* **230**, 105031 (2024).
2. The Guardian, Deaths and arrests in france after psg champions league victory (2025) Accessed: 2025-06-09.

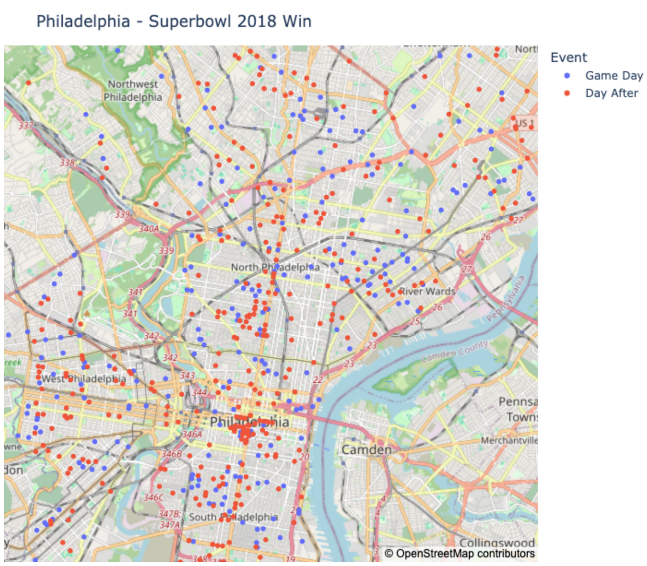
SI Appendix.

SI Appendix, A. Map Figures - For Location Analysis. The included graphs are from the event-based analysis in Chicago and Philadelphia.

(a) 2022 World Series Loss - Philadelphia



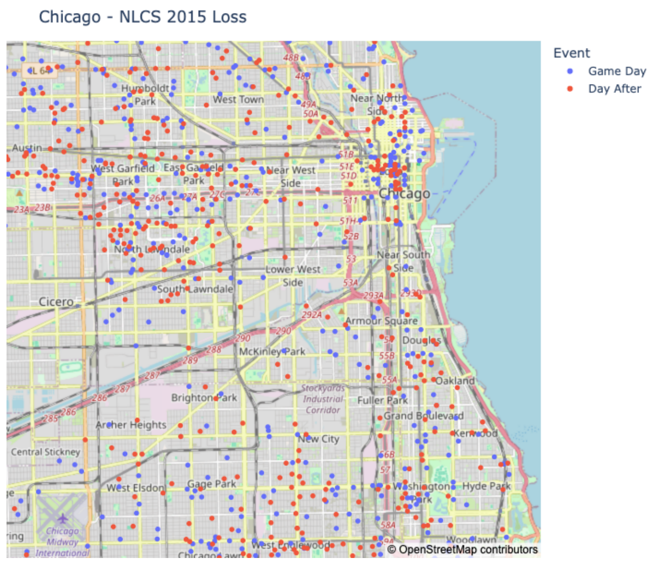
(b) 2025 Super Bowl Victory - Philadelphia



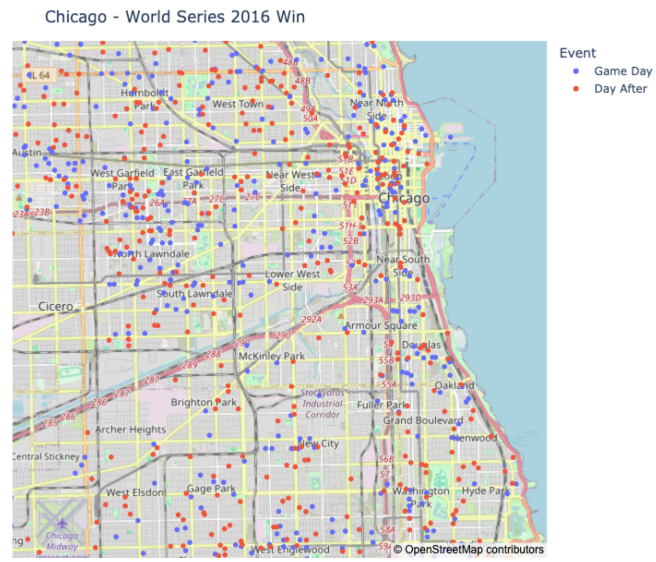
(c) 2018 Super Bowl Victory - Philadelphia

(d) 2008 World Series Victory - Philadelphia

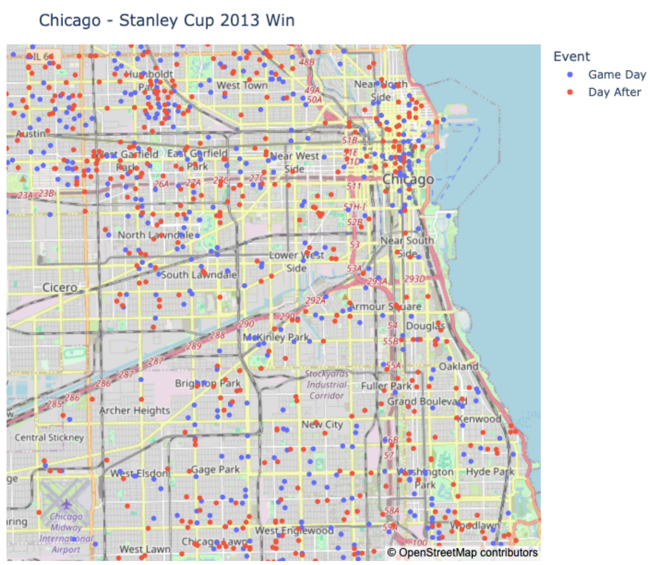
Fig. S1. Crime maps for major sporting events - Part 1. (a) Shows dispersed crime patterns with no significant spike following the 2022 World Series loss. (b) and (c) demonstrate concentrated crime clusters the day after Super Bowl victories in 2025 and 2018, respectively. (d) Shows crime patterns after the 2008 Phillies World Series victory.



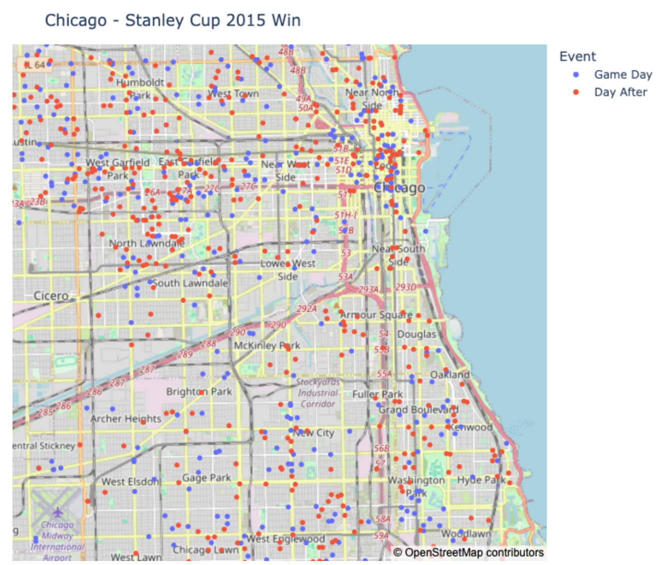
(a) 2015 NLCS - Chicago Cubs



(b) 2016 World Series Victory - Chicago Cubs



(c) 2013 Stanley Cup Victory - Chicago Blackhawks



(d) 2015 Stanley Cup Victory - Chicago Blackhawks

Fig. S2. Crime maps for major sporting events - Part 2. (a) Shows crime patterns during the 2015 Cubs NLCS. (b) demonstrates the 2016 Cubs World Series victory crime patterns. (c) and (d) show Chicago Blackhawks Stanley Cup victories in 2013 and 2015, respectively.